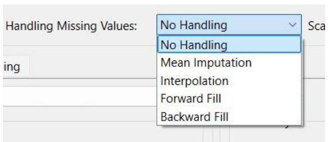
Machine Learning Gui

Mustafa Yarkin Konma

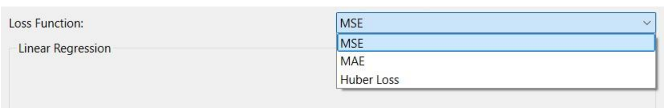
2206A605

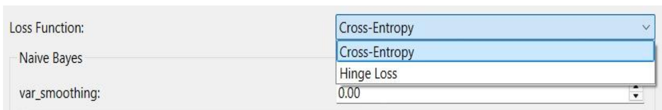
I have added Missing Value Handling as shown in the picture with its code

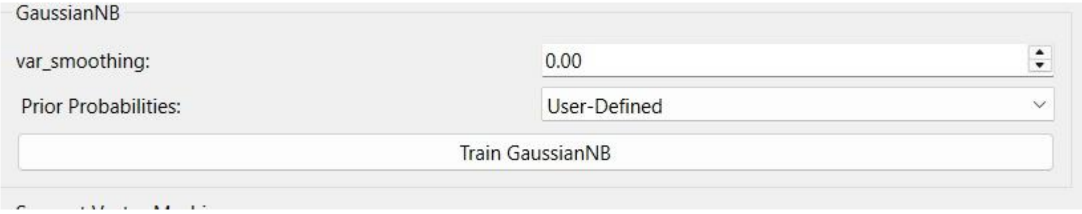
```
def handle_missing_values(self):
    """Apply selected missing data handling method."""
    method = self.missing_values_combo.currentText()

    if method == "No Handling":
        return

    elif method == "Mean Imputation":
        imputer = SimpleImputer(strategy="mean")

    elif method == "Interpolation":
        self.X_train = pd.DataFrame(self.X_train).interpolate().values
        self.X_test = pd.DataFrame(self.X_test).interpolate().values
        return

    elif method == "Forward Fill":
        self.X_train = pd.DataFrame(self.X_train).fillna(method="ffill").values
        self.X_test = pd.DataFrame(self.X_test).fillna(method="ffill").values
        return

    elif method == "Backward Fill":
        self.X_train = pd.DataFrame(self.X_train).fillna(method="bfill").values
        self.X_test = pd.DataFrame(self.X_test).fillna(method="bfill").values
        return
```



I have added Loss option to both classification and regression



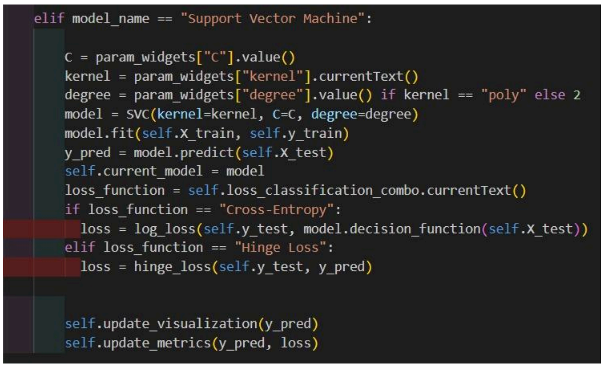




GaussianNB with var smoothing and prior probabilities

I have added Svm Modeling but for some reason it doesn't work

The code in picture



```
elif model_name == "Support Vector Machine":

    C = param_widgets["C"].value()
    kernel = param_widgets["kernel"].currentText()
    degree = param_widgets["degree"].value() if kernel == "poly" else 2
    model = SVC(kernel=kernel, C=C, degree=degree)
    model.fit(self.X_train, self.y_train)
    y_pred = model.predict(self.X_test)
    self.current_model = model
    loss_function = self.loss_classification_combo.currentText()
    if loss_function == "Cross-Entropy":
      loss = log_loss(self.y_test, model.decision_function(self.X_test))
    elif loss_function == "Hinge Loss":
      loss = hinge_loss(self.y_test, y_pred)


    self.update_visualization(y_pred)
    self.update_metrics(y_pred, loss)
```